



BOSagora - Token and Token Swap

Security Assessment

CertiK Assessed on May 13th, 2025





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BOSagora - Token and Token Swap

The security assessment was prepared by CertiK, the leader in Web3.0 security.

Executive Summary

TYPES

DeFi

ECOSYSTEM

Ethereum (ETH)

METHODS

Formal Verification, Manual Review, Static Analysis

LANGUAGE

Solidity

TIMELINE

Delivered on 05/13/2025

KEY COMPONENTS

N/A

CODEBASE

[BOSAGORA](#)

[View All in Codebase Page](#)

COMMITTS

- [0x63a3ea01e683aee93d0fb6ff93b35691230113fc](#)
- [0x51bd4f39803fcaeff3ef45aae2c3aaff0b9fadcb](#)

[View All in Codebase Page](#)

Vulnerability Summary



5

Total Findings

5

Resolved

0

Partially Resolved

0

Acknowledged

0

Declined

2 Centralization

2 Removed



Centralization findings highlight privileged roles & functions and their capabilities, or instances where the project takes custody of users' assets.

0 Critical

Critical risks are those that impact the safe functioning of a platform and must be addressed before launch. Users should not invest in any project with outstanding critical risks.

0 Major

Major risks may include logical errors that, under specific circumstances, could result in fund losses or loss of project control.

0 Medium

Medium risks may not pose a direct risk to users' funds, but they can affect the overall functioning of a platform.

3 Minor

3 Resolved



Minor risks can be any of the above, but on a smaller scale. They generally do not compromise the overall integrity of the project, but they may be less efficient than other solutions.

0 Informational

Informational errors are often recommendations to improve the style of the code or certain operations to fall within industry best practices. They usually do not affect the overall functioning of the code.

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I **Disclaimer**

CODEBASE | BOSAGORA - TOKEN AND TOKEN SWAP

Repository

[BOSAGORA](#)

Commit

- [0x63a3ea01e683aee93d0fb6ff93b35691230113fc](#)
- [0x51bd4f39803fcaeff3ef45aae2c3aaff0b9fadcb](#)

AUDIT SCOPE | BOSAGORA - TOKEN AND TOKEN SWAP

4 files audited ● 3 files without findings ● 1 file with Removed findings

ID	Repo	File	SHA256 Checksum
● BOA	bosagora/bosagora-erc20	 BOSAGORA.sol	6afb36371b648b4c5046f8155a1802ce0a4415f026340b68a358649266428cd0
● TSU	bosagora/bosagora-erc20	 TokenSwap.sol	427420770222181c619e66169031be0f71d68352bcc75704bf64e8edffb65b56
● TST	mainnet	 boa-contracts/contracts/TokenSwap.sol	0c4bdf3518a9baea265f2466747da00a2b6d3996b4617b0ef3a37b163781ac6
● BOG	mainnet	 boa-contracts/contracts/BOSAGORA.sol	1888c1675021efc4be499b6d06f97d927bf17b7e153c0b3fd8c076388e10a026

APPROACH & METHODS | BOSAGORA - TOKEN AND TOKEN SWAP

This report has been prepared for BOSagora to discover issues and vulnerabilities in the source code of the BOSagora - Token and Token Swap project as well as any contract dependencies that were not part of an officially recognized library. A comprehensive examination has been performed, utilizing Formal Verification, Manual Review, and Static Analysis techniques.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.

The security assessment resulted in findings that ranged from critical to informational. We recommend addressing these findings to ensure a high level of security standards and industry practices. We suggest recommendations that could better serve the project from the security perspective:

- Testing the smart contracts against both common and uncommon attack vectors;
- Enhance general coding practices for better structures of source codes;
- Add enough unit tests to cover the possible use cases;
- Provide more comments per each function for readability, especially contracts that are verified in public;
- Provide more transparency on privileged activities once the protocol is live.

REVIEW NOTES | BOSAGORA - TOKEN AND TOKEN SWAP

Overview

The **BOSagora** project involves the following smart contracts:

- **BOSAGORA**: This ERC20 token contract mints tokens gradually according to a predefined 128-year schedule, with cumulative yearly supply limits and fixed deadlines. Tokens are minted to a designated asset account whenever the current time exceeds the next scheduled deadline and the current supply is below the required limit.
- **TokenSwap**: A contract that enables users to swap their old tokens for new tokens on a 1:1 basis, with the old tokens being burned by sending them to a designated burn address.

External Dependencies

The following are external addresses used within the contracts:

BOSAGORA.sol

- `assetAccount`

We assume these contracts or addresses are valid and non-vulnerable actors and implement proper logic to collaborate with the current project.

FINDINGS | BOSAGORA - TOKEN AND TOKEN SWAP



5

Total Findings

0

Critical

2

Centralization

0

Major

0

Medium

3

Minor

0

Informational

This report has been prepared to discover issues and vulnerabilities for BOSagora - Token and Token Swap. Through this audit, we have uncovered 5 issues ranging from different severity levels. Utilizing the techniques of Formal Verification, Manual Review & Static Analysis to complement rigorous manual code reviews, we discovered the following findings:

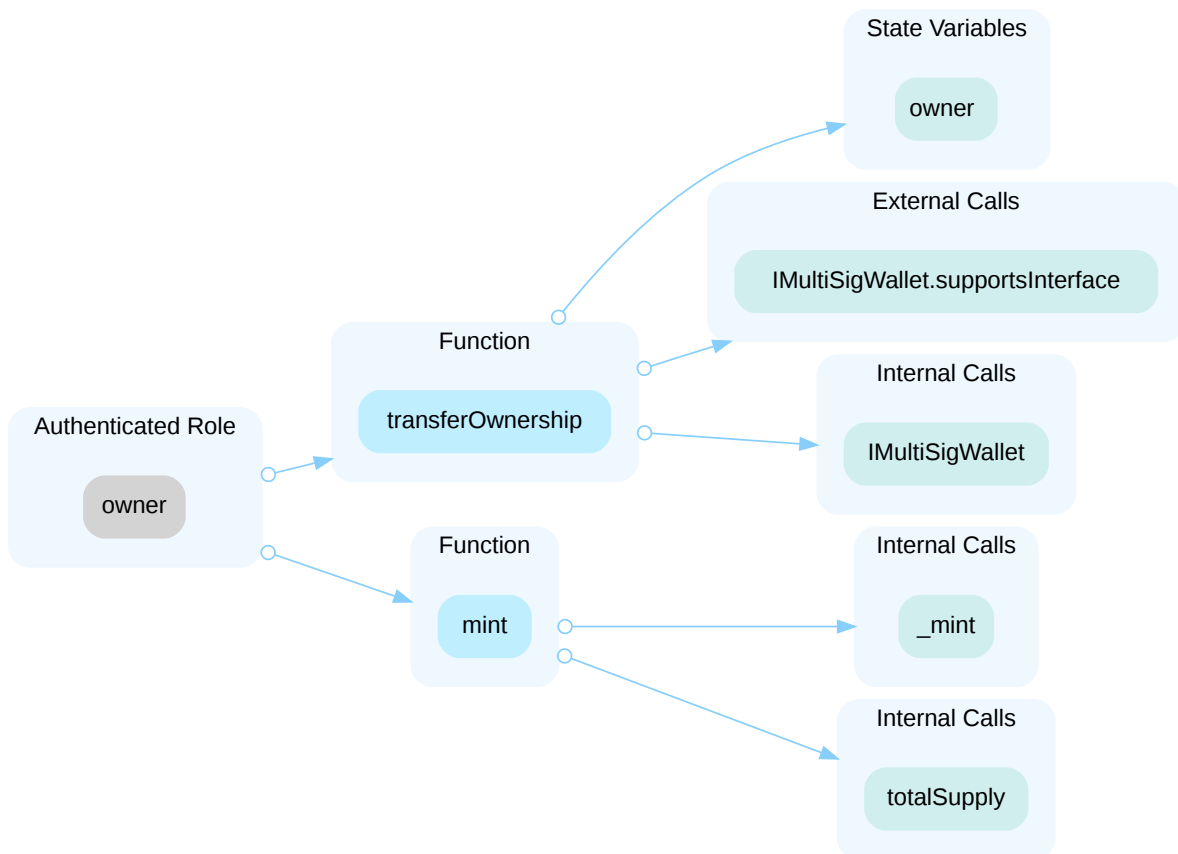
ID	Title	Category	Severity	Status
BTA-03	Centralization Related Risks	Centralization	Centralization	● Removed
BTA-04	Initial Token Distribution	Centralization	Centralization	● Removed
BTA-06	Incompatible With Non-Standard ERC20 Token (Ethereum USDT)	Logical Issue	Minor	● Resolved
BTA-07	Incompatibility With Deflationary Tokens (Non-Standard ERC20 Token)	Logical Issue	Minor	● Resolved
BTA-08	Lack Of Input Validation	Volatile Code	Minor	● Resolved

BTA-03 | CENTRALIZATION RELATED RISKS

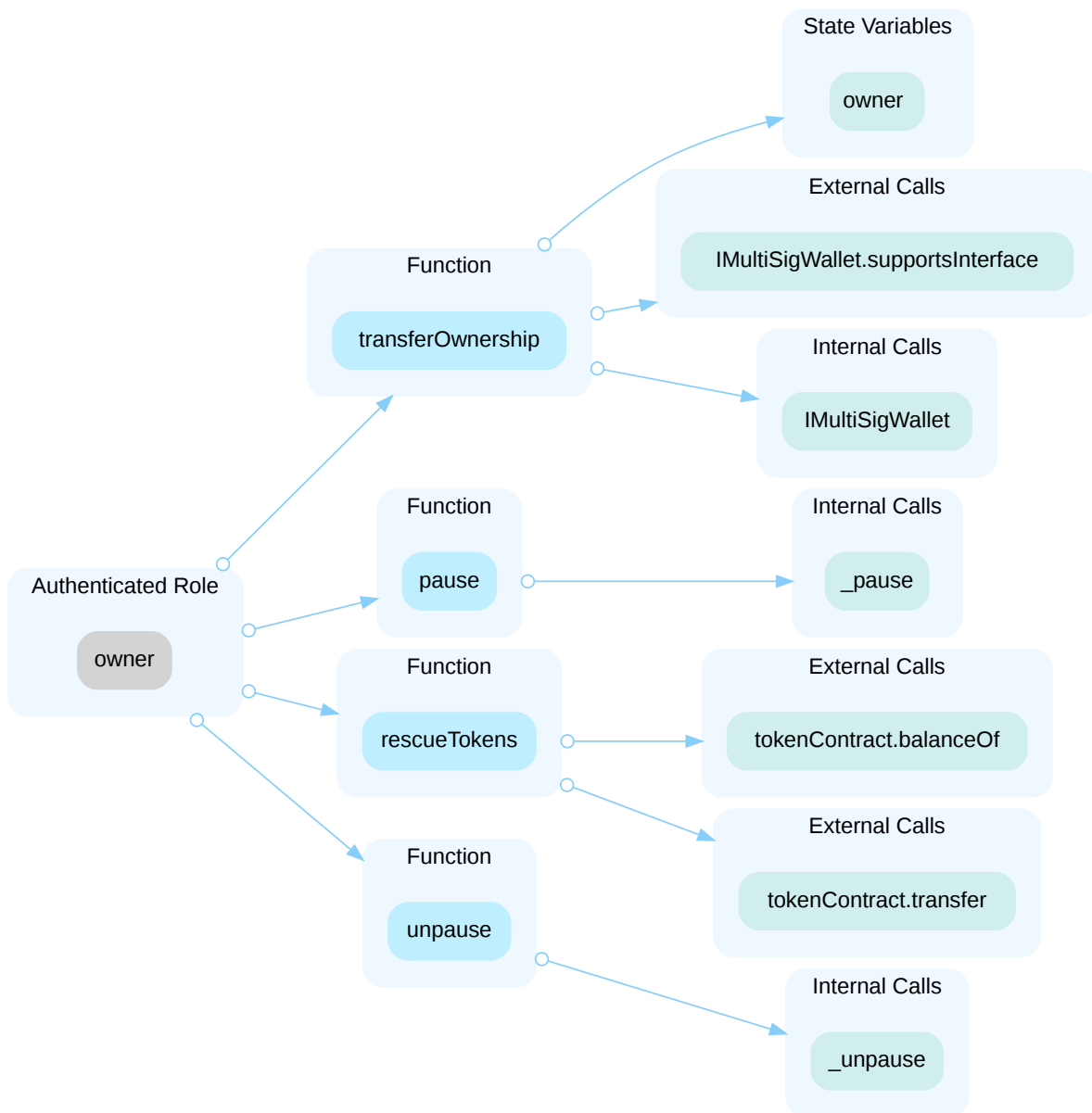
Category	Severity	Location	Status
Centralization	● Centralization	BOSAGORA.sol (fa18989): 79, 101; TokenSwap.sol (fa18989): 104, 133, 141, 200	● removed

Description

In the contract `BOSAGORA`, the role `owner` has authority over the functions shown in the diagram below. Any compromise to the `owner` account may allow the hacker to take advantage of this authority and transfer contract ownership to a new address or mint new tokens if conditions are met.



In the contract `TokenSwap`, the role `owner` has authority over the functions shown in the diagram below. Any compromise to the `owner` account may allow the hacker to take advantage of this authority and transfer ownership to a new owner address, pause the contract, rescue specified tokens to a designated address, and unpause the contract.



Recommendation

The risk describes the current project design and potentially makes iterations to improve in the security operation and level of decentralization, which in most cases cannot be resolved entirely at the present stage. We advise the client to carefully manage the privileged account's private key to avoid any potential risks of being hacked. In general, we strongly recommend centralized privileges or roles in the protocol be improved via a decentralized mechanism or smart-contract-based accounts with enhanced security practices, e.g., multisignature wallets. Indicatively, here are some feasible suggestions that would also mitigate the potential risk at a different level in terms of short-term, long-term and permanent:

Short Term:

Timelock and Multi sign (2/3, 3/5) combination *mitigate* by delaying the sensitive operation and avoiding a single point of key management failure.

- Time-lock with reasonable latency, e.g., 48 hours, for awareness on privileged operations;
AND
- Assignment of privileged roles to multi-signature wallets to prevent a single point of failure due to the private key compromised;
AND
- A medium/blog link for sharing the timelock contract and multi-signers addresses information with the public audience.

Long Term:

Timelock and DAO, the combination, *mitigate* by applying decentralization and transparency.

- Time-lock with reasonable latency, e.g., 48 hours, for awareness on privileged operations;
AND
- Introduction of a DAO/governance/voting module to increase transparency and user involvement.
AND
- A medium/blog link for sharing the timelock contract, multi-signers addresses, and DAO information with the public audience.

Permanent:

Renouncing the ownership or removing the function can be considered *fully resolved*.

- Renounce the ownership and never claim back the privileged roles.
OR
- Remove the risky functionality.

Alleviation

[BOSagora Team, 05/08/2025]:

We have implemented significant changes to address the centralization risks (BTA-03) identified in the previous audit. Here's how we've completely resolved these issues:

1. Removal of Privileged Accounts:
 - Removed all privileged accounts and ownership controls
 - Removed all administrative functions
 - Implemented fully permissionless design

2. BOSAGORA.sol Changes:

- Removed all privileged functions
- Implemented permissionless minting that anyone can trigger
- Added 128-year token distribution schedule
- All minted tokens are automatically sent to the multi-signature wallet (assetAccount)

3. TokenSwap.sol Changes:

- Removed all privileged functions
- Implemented immutable contract design
- Made contract fully immutable after deployment

4. Contract Addresses:

- BOSAGORA Token: 0x51bD4f39803fcAEFf3ef45aae2C3aaF0B9faDcb
- TokenSwap Contract: 0x63a3eA01E683aee93d0fb6fF93B35691230113fC

5. Multi-signature Wallet Information:

- Multi-sig wallet address: 0x41D2b13083f3b6d1Ba871eb3A61f0fe6A4d198b4
- Signers:
 - 0x2A1317CfC9fB4499F84Af1071eB44CfB43843eE2
 - 0x39E21BeeEB3d93E5F274aA70861cdE869b2AA4C7
 - 0x28c70b588C4ba0502fB9aD7Dac506039D320e07B
- Requires 2 out of 3 signatures for any transaction

These changes completely resolve the BTA-03 findings by:

- Removing all privileged accounts and functions
- Implementing fully permissionless design
- Eliminating the need for timelock and multi-sig controls
- Making contracts immutable after deployment
- Ensuring transparent and predictable token distribution
- Using multi-signature wallet for secure storage of minted tokens

The contracts have been deployed to Ethereum mainnet and are now fully operational. We request the audit team to review these changes and confirm that the centralization risks have been completely resolved.

[CertiK, 05/12/2025]:

The team redeployed the BOSAGORA and TokenSwap contracts, ensuring that all privileged functions were removed.

BTA-04 | INITIAL TOKEN DISTRIBUTION

Category	Severity	Location	Status
Centralization	● Centralization	boa-contracts/contracts/BOSAGORA.sol (05/05-BOSAGORA): 358	● removed

Description

The `BOSAGORA` token has a total supply of 4.95 billion and releases a certain number of tokens annually over a span of 128 years. All released tokens are minted to the `assetAccount` wallet. This poses a centralization risk because the anonymous account can distribute tokens without the community's consensus. If the deployer account holding the undistributed tokens is compromised, an attacker could potentially steal and sell the tokens on the market, causing significant harm to the project.

Recommendation

It is recommended that the team be transparent regarding the initial token distribution process. The token distribution plan should be published in a public location that the community can access. The team should make efforts to restrict access to the private keys of the deployer account or EOAs. A multi-signature (2/3, 3/5) wallet can be used to prevent a single point of failure due to a private key compromise. Additionally, the team can lock up a portion of tokens, release them with a vesting schedule for long-term success, and deanonymize the project team with a third-party KYC provider to create greater accountability.

Alleviation

[BOSagora Team, 05/05/2025]:

We have implemented significant changes to address the token distribution and centralization concerns raised in the previous audit:

1. Token Distribution Schedule Implementation:

- Implemented the 128-year token distribution schedule as specified in our whitepaper
- Added yearly deadline timestamps and cumulative supply limits for each period
- The minting process is now automated and follows a predetermined schedule
- Maximum supply is capped at 4.95 billion BOA tokens

2. Multi-signature Wallet Integration:

- All minted tokens are now sent to a multi-signature wallet
- Multi-sig wallet address: 0x41D2b13083f3b6d1Ba871eb3A61f0fe6A4d198b4
- Signers:
 - 0x2A1317CfC9fB4499F84Af1071eB44CfB43843eE2
 - 0x39E21BeeEB3d93E5F274aA70861cdE869b2AA4C7
 - 0x28c70b588C4ba0502fB9aD7Dac506039D320e07B
- Requires 2 out of 3 signatures for any transaction

3. Additional Security Measures:

- Implemented permissionless minting trigger (anyone can call mint after conditions are met)
- No ownership or administrative functions after deployment
- Fixed maximum supply with 7 decimal places
- All minting parameters are immutable after deployment

These changes directly address the BTA-04 findings by:

- Implementing a transparent and predictable token distribution schedule
- Removing single-point-of-failure risks through multi-signature implementation
- Ensuring token distribution follows the community-approved whitepaper
- Eliminating the possibility of unauthorized token minting

The contract has been deployed at: 0x51bD4f39803fcAEff3ef45aae2C3aaFf0B9faDcb

Token distribution plan is available at: https://bosagora.io/files/New_BOA_Token_Issuance_Plan_20250501.pdf

[CertiK, 05/12/2025]:

The team has redeployed the **BOSAGORA** contract, with tokens to be released annually to the 2/3 multi-signature wallet, following the outlined distribution plan.

Although this strategy has significantly reduced the risk, it is important to recognize that it has not entirely eliminated it. CertiK strongly recommends that the project team periodically review the private key security management for all wallet addresses

configured within the multi-signature wallets.

BTA-06 | INCOMPATIBLE WITH NON-STANDARD ERC20 TOKEN (ETHEREUM USDT)

Category	Severity	Location	Status
Logical Issue	Minor	TokenSwap.sol (fa18989): 167, 173, 176, 181, 184, 208	Resolved

Description

Non-standard ERC20 tokens, such as USDT on the Ethereum mainnet, do not return a value from their `transfer` and `transferFrom` functions as specified by the ERC20 standard. If the token being used is USDT, any attempt to execute these functions expecting a return value will cause the transaction to revert.

Recommendation

Since the token accepted by the project is not deterministic at the moment, we recommend using `SafeERC20.safeTransfer()` and `SafeERC20.safeTransferFrom()` methods in OpenZeppelin's `SafeERC20.sol` instead.

Alleviation

[BOSagora, 04/16/2025]:

We already only deal with ERC20 standards.

It will only exchange tokens that we issued.

We will only use this in exchange from the old BOA to the new BOA

[CertiK, 04/17/2025]:

The team confirmed that the contract exclusively supports the BOA token, which follows the standard ERC20 protocol.

[CertiK, 04/25/2025]:

Since the contract is deployed on the Ethereum mainnet, we'd like to inform the team that any USDT accidentally transferred to the contract will be permanently locked and cannot be recovered.

BTA-07 | INCOMPATIBILITY WITH DEFLATIONARY TOKENS (NON-STANDARD ERC20 TOKEN)

Category	Severity	Location	Status
Logical Issue	Minor	TokenSwap.sol (fa18989): 181	Resolved

Description

The project design may not be compatible with certain non-standard ERC20 tokens, such as **deflationary tokens** (fee-on-transfer tokens).

The functions use `transferFrom()` / `transfer()` to move funds from the sender to the recipient but fail to verify if the received token amount matches the transferred amount. This could pose an issue with fee-on-transfer tokens, where the post-transfer balance might be less than anticipated, leading to balance inconsistencies. There might be subsequent checks for a second transfer, but an attacker might exploit leftover funds (such as those accidentally sent by another user) to gain unjustified credit.

In the `swap` function of `TokenSwap` contract, if the old token has transfer fees, transferring the same amount from the current contract to the `BURN_ADDRESS` may cause a revert.

Scenario

When transferring deflationary ERC20 tokens, the input amount may not equal the received amount due to the charged transaction fee. For example, if a user sends 100 deflationary tokens (with a 10% transaction fee), only 90 tokens actually arrive to the contract. However, a failure to discount such fees may allow the same user to withdraw 100 tokens from the contract, which causes the contract to lose 10 tokens in such a transaction.

Recommendation

We advise the client to regulate the set of tokens supported and add necessary mitigation mechanisms to keep track of accurate balances if there is a need to support non-standard ERC20 tokens.

Alleviation

[BOSagora, 04/16/2025]:

We already only deal with ERC20 standards.

It will only exchange tokens that we issued.

We will only use this in exchange from the old BOA to the new BOA

[CertiK, 04/17/2025]:

The team confirmed that the contract only supports BOA token which is standard ERC20.

BTA-08 | LACK OF INPUT VALIDATION

Category	Severity	Location	Status
Volatile Code	Minor	TokenSwap.sol (fa18989): 104	Resolved

Description

In the `transferOwnership` function of the `TokenSwap` contract, the input parameter `newOwner` (the address of the new owner) is **not checked to ensure it is different from the current `owner`**. This means that calling `transferOwnership` with the same address as the current `owner` will execute successfully, even though it is effectively a redundant operation.

```
function transferOwnership(address newOwner) external onlyOwner {
    require(newOwner != address(0), "TokenSwap: New owner is zero address");
    require(

IMultiSigWallet(newOwner).supportsInterface(type(IMultiSigWallet).interfaceId),
    "TokenSwap: Invalid interface ID of new multi sig wallet"
    );

    address previousOwner = owner;
    owner = newOwner;
    emit OwnershipTransferred(previousOwner, newOwner);
}
```

- The function ensures:
 - The `newOwner` is not the zero address (`require(newOwner != address(0))`).
 - The `newOwner` implements the `IMultiSigWallet` interface.

However, it does **not check whether `newOwner` is the same as the current `owner`**, leading to unnecessary state updates and event emissions.

Recommendation

It's recommended to add a check to ensure that `newOwner` is not equal to the current `owner`.

Alleviation

[BOSagora, 04/16/2025]:

The team heeded the advice to fix the issue and changes were reflected in commit [2924d2a91aa0a5d19e25acba76853bc0b62755d1](#).

OPTIMIZATIONS | BOSAGORA - TOKEN AND TOKEN SWAP

ID	Title	Category	Severity	Status
<u>BTA-01</u>	Unnecessary Gas Waste On Reverting Transactions	Gas Optimization	Optimization	● Resolved
<u>BTA-02</u>	Redundant Transfers For Burning Old Tokens	Code Optimization	Optimization	● Resolved

BTA-01 | UNNECESSARY GAS WASTE ON REVERTING TRANSACTIONS

Category	Severity	Location	Status
Gas Optimization	● Optimization	TokenSwap.sol (fa18989): 176~177, 184~185	● Resolved

Description

In the following part of the `swap` function of `TokenSwap` contract:

```
if (!newTokenTransferSuccess) {  
    // If new token transfer fails, return old token to user  
    require(oldToken.transfer(msg.sender, amount), "TokenSwap: Failed to  
return old token");  
    revert("TokenSwap: New token transfer failed");  
}
```

```
if (!burnSuccess) {  
    // If burning fails, return old token to user  
    require(oldToken.transfer(msg.sender, amount), "TokenSwap: Failed to  
return old token");  
    revert("TokenSwap: Burning old token failed");  
}
```

If `newTokenTransferSuccess` or `burnSuccess` is `false`, `oldToken` is returned to the user before reverting the transaction. However, since the entire transaction will revert, the state changes (including the `transfer` of `oldToken` back to the user) will be reverted as well. This means the `transfer` operation is unnecessary and wastes gas.

Recommendation

It's recommended to simply revert the transaction directly when `newTokenTransferSuccess` or `burnSuccess` fails. This eliminates the gas cost of the unnecessary `transfer` operation.

Alleviation

[BOSagora, 04/16/2025]:

The team heeded the advice to fix the issue and changes were reflected in commit

[b1c800db03fab30a6a38baa5708f565537a6a36](#).

BTA-02 | REDUNDANT TRANSFERS FOR BURNING OLD TOKENS

Category	Severity	Location	Status
Code Optimization	● Optimization	TokenSwap.sol (fa18989): 181~186	● Resolved

Description

The `swap` function first transfers the `oldToken` from the user to the contract, and then transfers it from the contract to the `BURN_ADDRESS`:

```
bool oldTokenTransferSuccess = oldToken.transferFrom(msg.sender,
address(this), amount);
if (!oldTokenTransferSuccess) {
    revert("TokenSwap: Old token transfer failed");
}
...
bool burnSuccess = oldToken.transfer(BURN_ADDRESS, amount);
if (!burnSuccess) {
    // If burning fails, return old token to user
    require(oldToken.transfer(msg.sender, amount), "TokenSwap: Failed to
return old token");
    revert("TokenSwap: Burning old token failed");
}
```

This involves two separate `transfer` operations:

1. From user to contract: `transferFrom`.
2. From contract to `BURN_ADDRESS`: `transfer`.

Each `transfer` operation costs gas, and the second `transfer` is redundant since we can directly send the tokens to the `BURN_ADDRESS` during the initial `transferFrom`.

Recommendation

It's recommended to transfer old tokens from the user to the `BURN_ADDRESS` using `transferFrom`. This eliminates the second `transfer` operation, reducing gas consumption.

Alleviation

[BOSagora, 04/16/2025]:

The team heeded the advice to fix the issue and changes were reflected in commit

[9db35cb887fb3f5d5e7665387fde741d667392c9](#).

APPENDIX | BOSAGORA - TOKEN AND TOKEN SWAP

Finding Categories

Categories	Description
Gas Optimization	Gas Optimization findings do not affect the functionality of the code but generate different, more optimal EVM opcodes resulting in a reduction on the total gas cost of a transaction.
Volatile Code	Volatile Code findings refer to segments of code that behave unexpectedly on certain edge cases and may result in vulnerabilities.
Logical Issue	Logical Issue findings indicate general implementation issues related to the program logic.
Centralization	Centralization findings detail the design choices of designating privileged roles or other centralized controls over the code.

Checksum Calculation Method

The "Checksum" field in the "Audit Scope" section is calculated as the SHA-256 (Secure Hash Algorithm 2 with digest size of 256 bits) digest of the content of each file hosted in the listed source repository under the specified commit.

The result is hexadecimal encoded and is the same as the output of the Linux "sha256sum" command against the target file.

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